

a single intervention [27]. This evolution has enabled more personalized and comprehensive adherence support. However, because many interventions now combine multiple components, it remains unclear which behavioral strategies contribute most to improved medication adherence [28]. As such, this scoping review synthesizes recent evidence on digital behavioral interventions designed to improve medication adherence in people with T2D by mapping intervention characteristics, identifying their behavioral components, summarizing approaches used to assess medication adherence, and describing reported glycemic outcomes. By providing an updated overview of contemporary digital adherence interventions, this review aims to inform future research and support the development of more effective and patient-centered strategies to improve medication adherence in diabetes care.

Methods

A scoping review methodology was selected because of the diversity in study designs, intervention approaches, and outcome measures in this field [29]. The review followed the framework proposed by Arksey and O'Malley, which includes identifying the research question, identifying relevant studies, study selection, data charting, and summarizing the results [30]. Reporting of the review was guided by the Preferred Reporting Items for Systematic Reviews and Meta Analyses Extension for Scoping Reviews to enhance transparency and reproducibility [31]. A review protocol was developed a priori to guide the review process, including the research question, eligibility criteria, search strategy, data extraction procedures, and analytic approach. The protocol was used internally by the research team to ensure methodological consistency but was not formally registered or made publicly available.

Identifying the Research Questions

The research question was structured using the Population, Concept, and Context framework, which is commonly applied in scoping reviews synthesizing health services and health promotion research [32]. The population included individuals with T2D. The concept focused on digital technology driven educational and/or behavioral interventions aimed at improving medication adherence. The context included clinical and/or community settings in which such interventions were implemented (Table 1). The review aimed to identify recent digital interventions targeting medication adherence and to map the key components commonly used within these programs.

Information Sources and Search Strategy

A literature search was conducted in PubMed, Scopus, and Web of Science to identify relevant studies published between January 2021 and December 2025. The search strategy combined terms related to T2D, medication adherence, education, and behavioral or digital interventions. Keywords included “type 2 diabetes,” “medication adherence,” “medicine adherence,” “education,” and “intervention,” which were combined using Boolean operators. Searches were limited to titles and abstracts to improve specificity. All records identified through the database searches were exported to EndNote for reference management and removal of duplicate records. The full search strategies are provided in Appendix 1.

Study Selection

Titles and abstracts were screened using predefined eligibility criteria (Table 1) by one reviewer and independently verified by a second reviewer. Full text articles were then assessed independently by two reviewers. Any disagreements

Table 1 Research question development and study eligibility criteria guided by the Population-Concept-Context (PCC) framework

PCC tool	Inclusion criteria	Exclusion criteria
P – Population/Problem	• People with type 2 diabetes	• Caregivers • Healthcare professionals
C - Concept	• Core components of intervention or educational strategies intended to support better medication adherence • Use of digital technology-driven educational or behavioral interventions to improve medication adherence, defined as the initiation, implementation, and persistence of prescribed diabetes medications	• Absence of digital technology-driven educational or behavioral interventions to improve medication adherence • Studies focusing exclusively on diet, physical activity, foot care, or glycemic control, without addressing medication adherence
C - Context	• Use of digital technologies in clinical or community settings	• Emergency or urgent care settings
Type of studies	• Qualitative, quantitative, or mixed methods studies • Observational and experimental, cross-sectional, or longitudinal, randomized controlled trial or nonrandomized or noncontrolled trial, case series or case reports	• Review, meta-analysis, protocol, conference paper, editorials, letters, notes, book chapters, erratum or retraction
Language	• English	• Language other than English
Publication date	• From January 2021 to 2025 December	